simStateSpace: Simulate Data from State Space Models

Ivan Jacob Agaloos Pesigan

Description

Provides a streamlined and user-friendly framework for simulating data in state space models, particularly when the number of subjects/units (n) exceeds one, a scenario commonly encountered in social and behavioral sciences. This package was designed to generate data for the simulations performed in Pesigan, Russell, and Chow (2025: https://doi.org/10.1037/met0000779).

Installation

You can install the CRAN release of `simStateSpace` with:

```
install.packages("simStateSpace")
```

You can install the development version of `simStateSpace` from GitHub with:

```
if (!require("pak")) install.packages("pak")
pak::pkg_install("jeksterslab/simStateSpace")
```

More Information

See GitHub Pages for package documentation.

Citation

To cite `simStateSpace` in publications, please cite Pesigan et al. (2025).

References

Pesigan, I. J. A., Russell, M. A., & Chow, S.-M. (2025). Inferences and effect sizes for direct, indirect, and total effects in continuous-time mediation models. *Psychological Methods*. https://doi.org/10.1037/met0000779

R Core Team. (2026). *R: A language and environment for statistical computing*. R Foundation for Statistical Computing. Vienna, Austria. https://www.R-project.org/